

Topic: Visualising relational and network data

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Unit: FIT5147 Data exploration and visualisation - S2 2026

Book: Topic: Visualising relational and network data

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1. Introduction

In this topic, you will learn about the main kinds of visualisations for showing abstract data such as networks, hierarchies, control and process flow, and set relationships.

We will also look at clustering and common statistical measures such as centrality and cohesion for understanding networks. These are an important part of the data scientist's toolbag. Originally developed for social networks they are useful in understanding the rich and complex interrelationships between more abstract data.



Exploring the learning materials

As you work through the learning materials in this Moodle book, please complete the learning activities and consider the reflection opportunities presented. Various videos and webpages may be included for your reference to reinforce and expand on the contents of this book. You are recommended to watch any videos produced by Monash University. Unless stated otherwise, the other resources are optional but you may find them very helpful. If any links no longer work, try searching for the title elsewhere.



Recommended readings

- Take a look at the TED Talk by Manuel Lima. [A visual history of human knowledge](#) (TED 2015) on the history of the tree and network visualisation contains some really beautiful visualisations.

This topic is based upon

- [Statistical Analysis of Network Data with R](#) (Kolaczyk and Csárdi 2014).
- [Multivariate Network Visualization](#) (Andreas, Purchase and Ward 2014).
- Chapter 9 of [Interactive data visualisation: foundations, techniques, and applications](#) (Ward, Grinstein and Keim 2015).
- Chapter 9 of [Visualization Analysis and Design](#) (Munzner 2014).

2. Kinds of networks

Watch the following video which discusses network data and different kinds of networks.

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

These captions have been automatically generated. As such, the quality is not 100% accurate. Accuracy may be affected by audio quality, topic under discussion and clarity of the speakers. Staff are not expected or required to edit or correct automatically-generated captions.

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Network data and kinds of networks (2022) Monash University <[Panopto Viewer](#)>

Resources

You might be looking for these resources referenced in the video.

Kinds of networks

- [50 great examples of data visualisation](#) (Web Designer Depot 2009).
- [Infographic: Game of Thrones family tree](#) (Time 2023).
- [Political communication Modi style: A case study of the demonetisation campaign on Twitter](#) (Rodriguez & Niemann 2019).
- [Athenahealth visualisation](#) (Fathom 2016).
- [Why do cats & dogs ...?](#) (Bremer n.d.).
- [100 years of rock](#) (Concerthotels.com n.d.).
- [Make interactive network visualisations without coding](#) (Flourish n.d.).

3. Network analysis and visualisation

Networks are well-suited to modelling abstract relationships between entities. *Social networks* (or *sociograms*) are one of the best-known kinds of networks, these model communities and show the strength of the connection between different people in the community. Networks are also common in software engineering, where they are used to show the structure of software or its execution, and in the life sciences, where they are used to model processes, for instance, metabolic pathways.

Please look at the TED Talk by Nicholas Christakis on the the hidden influence of social networks (21 min) to see great examples of social network analysis and visualisation.



Nicholas Christakis: The hidden influence of social networks (2010) TED <[TED page](#)>



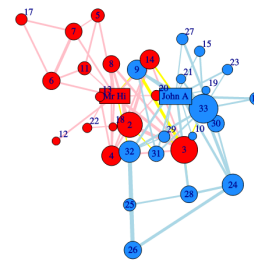
Apply it

Our running example in this topic will be the 'karate club network' of Zachary. This shows the amount of social interaction between karate club members during the time the club split into two clubs because of differences between one of the headteachers and the administrator.



Karate club network

And here is the network laid out to show the two factions (Mr Hi, '1' and John A, '34') with connections within factions shown in red or blue and inter-faction connections in yellow. The width of each edge is proportional to the strength of interaction, and the vertex's size is proportional to the total width of the edges entering it.



Two factions

Networks are becoming a much more common way of organising data because of the rise of graph databases such as [Neo4j](#) (Wikipedia 2023). While traditional relational databases support tabular data, graph databases support networks. Graph databases have the advantage that they scale up to larger data sets and are less rigid, allowing data to be stored in more ad hoc relationships, which can evolve.

4. Some network terminology

Networks are formalised as multivariate graphs or networks. These have *nodes* (also called *vertices*) which correspond to entities such as people or businesses, *edges* (sometimes called arcs) which connect two nodes and which represent a relationship such as 'is-an-employee', and *attributes* or *properties* of either nodes or edges such as 'years-employed' or 'age'. The edges are said to be *weighted* if they have strength or magnitude.

A network is *dynamic* if it changes over time.

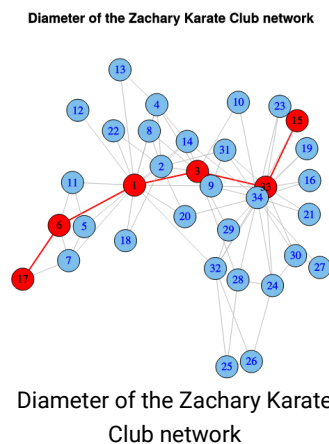
The network is *directed* if the edges have a direction: a diagram showing the inheritance relationship between classes in C++ or Java code is an example of a directed network. The karate club is an example of an undirected network.

A node's neighbours are the nodes connected to it by a single edge. A path between two nodes is a sequence of edges between the two of them. It is the *shortest path* if it has the least length. The shortest path length between two nodes is the (*graph-theoretic*) *distance* between them. In a social network, this is called the *degree of separation*. A network's diameter is the shortest path between any two nodes in the network. Two nodes are *connected* if there is a path between them, otherwise, they are *unconnected*. A path is *directed* if the edges in it are directed and flow from the start to the end of the path.



Apply it

The following diagram highlights the diameter of the Zachary Karate Club network, i.e., the longest shortest path between any two nodes in the network.



One apparent limitation of a network is that it can only represent binary relations. A network in which edges can connect more than two nodes is called a *hypergraph*, and the edges are called *hyperedges*. However, it turns out that for most practical purposes, hypergraphs are unnecessary as they can be modelled using a standard binary edge network by introducing a new node corresponding to each hyperedge and linking this node to each of the original nodes connected by the hyperedge using a binary edge.

5. Visual representations of networks

The most popular way of representing a network is using a *node-link diagram* like that showing the karate club social network. These place the nodes on the drawing canvas and use lines between them to represent edges. The edges are typically drawn as straight lines but can be curved edges, have multiple straight-line segments, or be drawn using only horizontal and vertical segments in the style of an electrical circuit.

Creating a good node-link diagram is not easy. Edges should not cross nodes, and a good diagram should try and reduce edge crossings, have straight edges, have roughly uniform edge lengths, and show symmetry and the graph structure. Unfortunately, these criteria conflict, so there is a trade-off between them. Furthermore, even optimising one of the criteria, such as reducing-edge crossings, is an NP-hard problem, so automatic network layout is computationally a very hard problem.

Fortunately, many different layout algorithms have been developed for node-link diagrams. The most common approach is the so-called *force-directed approaches*. Many are loosely based on a physical model of the network in which the nodes are spheres that repel each other, and the edges are springs with a natural length that connect the spheres. The layout is found by optimisation to find a placement for the nodes that minimises the total energy in the physical model. Closely related are *multi-dimensional scaling (MDS)* based approaches. In this approach, the distance between each pair of nodes is set to be the graph-theoretic distance between them: thus, if they are neighbours, the distance is 1. If there are N nodes, then this gives a distance between all of the nodes in N dimensional space. To find the network drawing, MDS techniques are used to project this space onto two-dimension so as find the placement that best preserves this graph-theoretic distance. The Kamada-Kawai algorithm is an MDS approach that uses a force-directed algorithm to find the positions. It was used for the above layouts.

Force-directed and MDS-based approaches place the nodes anywhere in the plane. Other drawing styles restrict placement options. In a *chord diagram*, the nodes are placed on the perimeter of a circle and connected by internal curved edges, while in an *arc diagram*, the nodes are placed on a line and connected by curved edges.

In the case of networks with directed edges, it is common to use some kind of *layered diagram* in which the edges are aligned in bands and edges that (generally) flow from left to right or from top-to-bottom. A variant is *radial network diagrams* in which the edges radiate outwards from a central node, and the layers are concentric circles.

Representing networks using some kind of node-link diagram is the most popular way of showing the network. Node-link diagrams have the advantage that most people are familiar with them, and for small networks, they support many of the abstract tasks that are important for network data, such as finding paths between nodes or understanding the structure. However, for larger networks, particularly those that are dense in the sense that they have lots of edges, edge crossings and general edge clutter make it difficult to understand the network structure and distinguish edges. While techniques like edge-bundling or grouping nodes with common edges in so-called power graphs reduce this clutter, node-link diagrams are very difficult to understand after a certain network size and node density.

The adjacency matrix representation is an alternative visualisation with the advantage of scaling to larger networks. If there are N nodes, then the matrix is $N \times N$, and the entry in the cell (i, j) if there is an edge from node i to node j . Of course, the matrix will be symmetric for undirected networks, so only one of the diagonal halves needs to be shown.

For the adjacency matrix representation to be effective, the rows and columns should be reordered to place connected nodes next to each other and to try and create rectangular 'blobs'. The row and column ordering should be consistent to preserve the reader's orientation. With this representation, tightly connected clusters can be seen as 'blobs' down the diagonal. The disadvantage of the adjacency matrix representation is that, at least for small graphs, it does not show the overall structure as clearly, it is difficult to find paths between nodes, and it is unfamiliar to most people. On the other hand, it has a great advantage that edge clutter is completely removed.

6. Visualising networks

Watch the following video, which discusses common ways to visualise network data (relational data).

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

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Visualising networks (2022) Monash University <[Panopto Viewer](#)>



Resources

You might be looking for these resources referenced in the video.

Common Ways to Visualise Network Data (Relational Data)

- [Datavizproject](#) (DVP n.d.).
- [From data to vis: Chord diagram](#) (From data to vis n.d.).

7. Different network diagrams

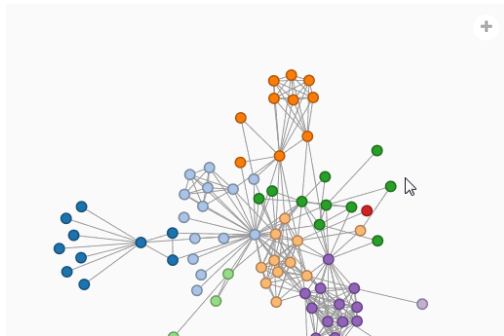
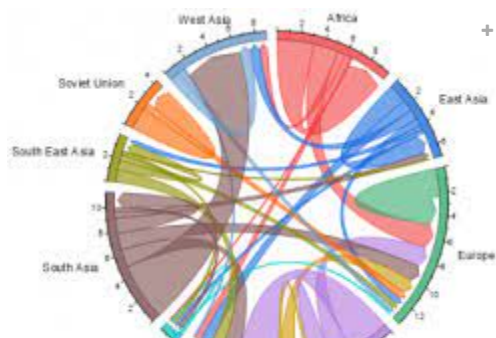
Purpose

This activity is intended to check your familiarity with different network diagrams.

Task

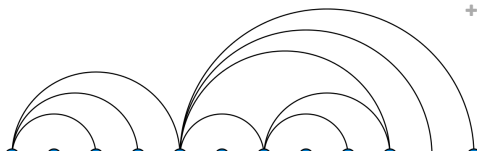
Step 1: Complete the following activity which asks you to identify the name of the network types.

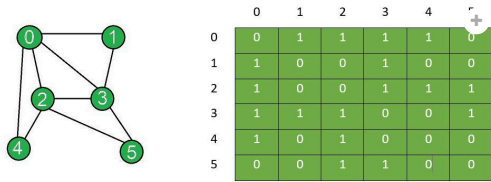
Select the correct network diagram name from the dropdown list to match the image of the visualisation shown. Use the  control to expand the image.



What is the name for this visualisation

What is the name for this visualisation





What is the name for this visualisation

What is the name for this visualisation

8. Common analytics for networks

There are several common statistical measurements for characterising the structure of the network and the importance of nodes and edges.

One fundamental measure is the *degree* of a node. This is the number of edges incident to the node (equivalently, the number of neighbours). We can plot the distribution of the node degree. In the case of weighted graphs, node degree is generalised to node strength which is the sum of the weights of the incident edges.

The frequency of node degree can give insight into the underlying graph. *Scale-free networks* are a common kind of network in which the degree distributions follows a power law. In other words, the number of nodes having degree k in the network is proportional to $k^{-\lambda}$ where λ is typically between 2 and 3. The network of citations between scientific papers forms a scale-free network, with some papers being massively cited.

Another interesting class of networks are the so-called *small-world networks*. In these networks, the average distance between two nodes grows logarithmically in the size of the network rather than linearly. Social networks are often claimed to be small-world networks as almost everyone has a friend or a friend of a friend in common. Many small-world networks are scale-free.

Centrality

A major focus in network analytics is to measure the importance or *centrality* of a node n . There are a variety of approaches:

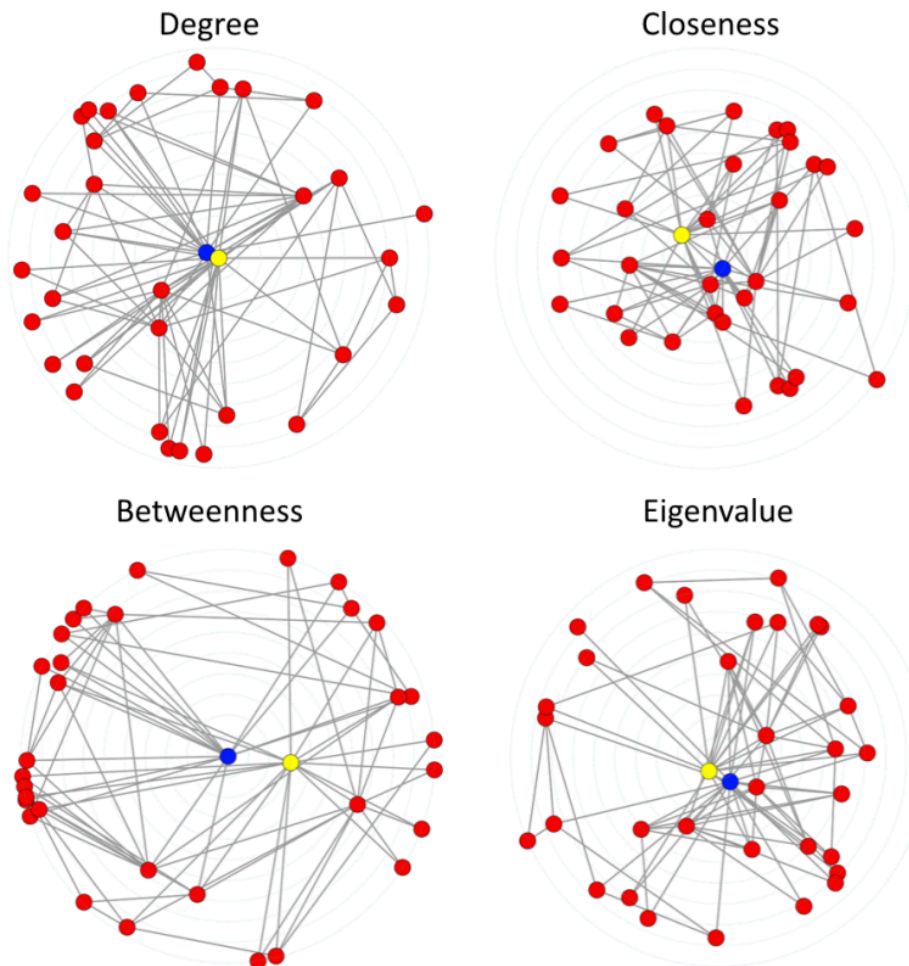
- The simplest measure is the *degree* of n : the higher the degree, the more important n is.
- *Closeness centrality* captures the idea that a node n is central if it is close to many other nodes: this measure varies inversely with the total of the node's graph-theoretic distance to all other nodes in the graph.
- *Betweenness centrality* summarises the extent to which the node n lies between the other nodes: it is based on counting the shortest paths that pass through n .
- Status or rank centrality is based on the assumption that a node is more important if it has important neighbours. These are found using eigenvalues and eigenvectors.

When exploring node centrality, it is common to use a node-link diagram in which nodes are placed radially with the nodes of most centrality at the centre.



Apply it

The following diagrams show different centrality metrics for the Karate Club Network



Different centrality metrics for the Karate Club Network (n.d.) adapted from Kolaczyk and Csárdi

We can also consider the centrality of an edge e . Here the most obvious measure is *betweenness centrality*: this is the number of shortest paths that pass through e .

Network cohesion

Another major focus in network analytics is *network cohesion*. This investigates how tightly connected the graph is.

- One approach is to measure the number of highly connected sub-graphs. A *clique* is a subset of nodes for which there is an edge between each pair of nodes in the subset, while a *bi-clique* is two subsets of nodes such that every node in each subset has an edge over the nodes in the other subset. One network analytic measures the number of different-sized cliques and bi-cliques in the network. Closely related is the *k-core*: a subset of nodes in which each node has at least k edges to other nodes in the subset.
- *Density* captures the frequency of actual edges compared with the total number of edges. The density of a subgraph is $\frac{M}{N(N-1)/2}$ where M is the number of edges and N is the number of nodes.
- The *clustering coefficient* measures the frequency in which sub-graphs with 3 connected nodes are actually cliques, i.e. there is an edge between all 3 nodes.
- *Vertex* and *edge connectivity* measure how many nodes or edges need to be removed from a connected network before it is disconnected.

Assortative mixing

Assortative mixing looks at how the attributes of a node affect the kinds of nodes it is linked to. For instance, in a social network, this might reveal that gender affects friendship and that men are likelier to be friends with men and women with women. If the nodes can be divided into K categories based on their attribute values, then assortative mixing computes the assortativity coefficient for each pair of categories. This ranges between -1 and 1. It is 0 if the category does not affect the frequency of edges and 1 if edges only occur between these two categories.

Graph clustering and partitioning

The final focus in network analytics is how to cluster the nodes in a graph. This is also called *community detection*. The aim is to partition the graph into subgraphs so that each subgraph is highly connected while the subgraphs are loosely connected.

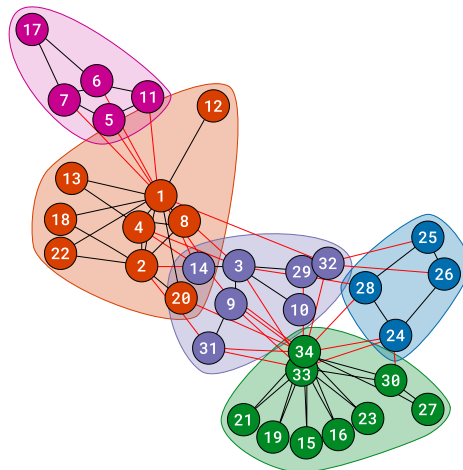
- *Hierarchical clustering* techniques can be used for this. One approach is to identify small highly connected subgraphs and then repeatedly merge the most highly connected subgraphs. This produces a hierarchy of nested partitions for which a dendrogram can show the hierarchy and the 'connectedness' between sub-components.
- *Spectral partitioning* uses eigenvalues to split the graph into smaller components recursively.

Clustering plays an important role when exploring networks, but the results should always be visualised to see if they make sense and also if analysis of the attributes etc., can help to explain the partitioning.



Apply it

The following diagram shows the Karate Club network data with clustering.



The "karate club network" with clustering (2022) <<https://bit.ly/3iFR59i>>

9. Analysing networks

Watch the following video, which discusses analysing networks, including the concepts of network centrality and network cohesion.

Select the advanced player features ^ control to access the video table with chapters targeting each topic.

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Analysing networks (2022) Monash University <[Panopto Viewer](#)>

10. Trees and hierarchies

Trees are a common kind of network in which there are no cycles, i.e., there is at most one path between each pair of nodes. If the edges are directed then the tree is called a hierarchy. In a true hierarchy for each node n there is at most one node m with an edge from m to n . m is called the parent of n . An organisation structure is often a hierarchy as is the subclass relationship when multiple inheritances are not allowed.

There are a number of quite different ways of drawing hierarchies. These rely on different visual or spatial relationships for showing the parent relationship. One of the most common ways is to use an indented list: file directories are often shown like this.

One example of a layered tree we have met throughout this unit is the [dendrogram](#) (Wikipedia 2022). They are widely used to show the similarity between hierarchical clusters with the vertical distance between parent and child proportional to their difference. [Cladograms](#) (Wikipedia 2022) are very similar to dendrograms but show the evolutionary history of related species.

However, using lines to show the parent relationship is not the only possible visual metaphor. Icicle and sunburst plots use adjacent overlap to indicate the parent relationship while treemaps use spatial containment. Treemaps are widely used to show hierarchically organised quantitative data, with the size of the rectangle being proportional to the data value. While they are visually appealing, we have found that using an indented list with an associated bar chart also works quite well and is easier to navigate around.

Containment is commonly combined with node-link representations when showing networks whose nodes have a hierarchical structure. This structure might be given or inferred by network clustering.

11. Flow diagrams

Networks are also used to show flow and processes. [Flow diagrams](#) (Wikipedia 2022) show the steps or events in a process. An example of flow diagrams is flow charts. These were once widely used to show control flow in algorithms and business processes.

Another kind of flow diagram is the [Sankey diagram](#) (Wikipedia 2024) which shows the amount of flow between different processes using variable width arrows or lines where the width is proportional to the amount of flow. They are similar to flow maps except that flow is between abstract entities rather than geographic locations.

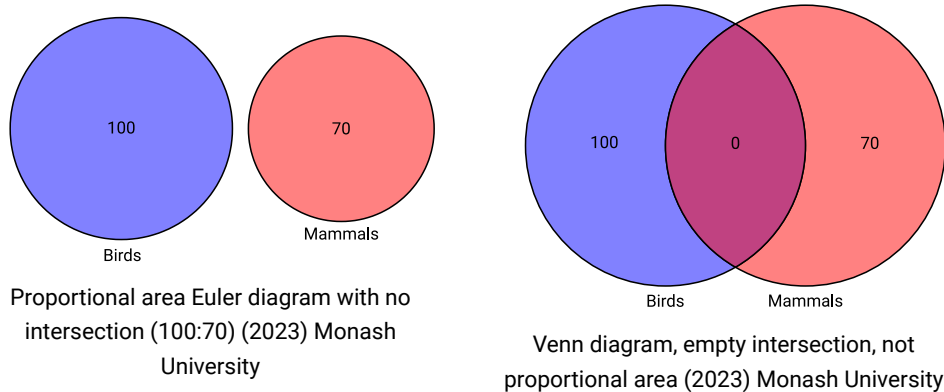
One problem with Sankey diagrams and other flow diagrams is that, like all node-link diagrams, the diagram becomes cluttered and confusing for large dense networks because of edge crossings and overlap. In such cases, matrix representations may be a better approach. Recent research in this area has investigated matrix representations like [MatrixWave](#) for showing abstract flow and [MapTrix](#) for showing flow between geographic locations.

12. Set visualisation

An important class of abstract relationships are based on set containment. Specialised diagrams have been developed to show the relationship between sets and their union and intersection. The best known are Venn and Euler diagrams. These use spatial containment to represent set containment.

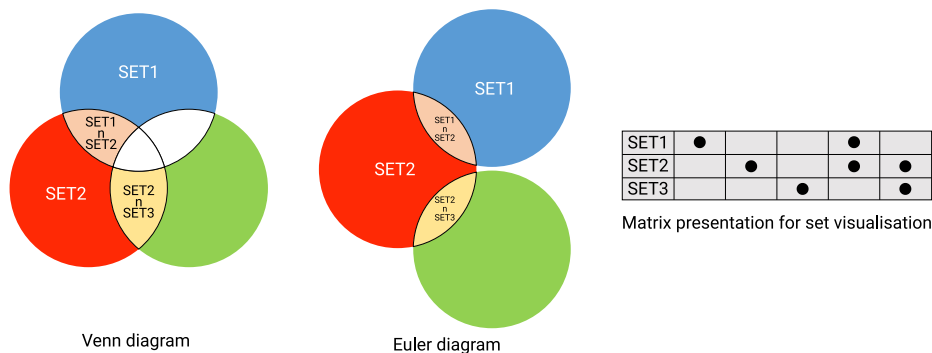
Venn diagrams differ from Euler diagrams in that the diagram must contain all possible set intersections, with shading showing which intersections are empty. Euler diagrams do not show empty set intersections. In the proportional area, Euler diagrams the size of the regions as proportional to the data quantity, such as the number of members or probability. These are used in medicine.

The following two diagrams were both created using the R library VennDiagram.



Like node-link representations for networks, Venn and Euler diagrams are intuitively easy to understand and allow small data sets to be readily understood. However once the number of sets increases they are difficult to draw and become extremely difficult to understand. Again a more abstract matrix representation provides a scalable alternative but one that is unfamiliar to most people. This matrix representation is the *linear diagram*. There is a row for each set and a column for each non-empty intersection of sets. The column width can be set to the size of this intersection.

Here are the same 3 sets drawn using these three different representations.



Venn diagram, Euler diagram and matrix presentation for set visualisation (2023) Monash University

13. Network Earth

Purpose

This activity is intended to demonstrate how network visualisations can be used to communicate a message about the real world.

Task

Step 1: Watch this video about natural networks - 'Network Earth' by Nature - which uses network visualisations to try to understand and predict the complexity of our multi-networked world.

Step 2: Consider the reflection questions which follow.

Network Earth

nature video



Watch on

Network Earth (2016) <<https://bit.ly/3Yw6adM>>



Reflect

These visualisations are helping us to see how the different elements in the network relate to each other. The transition between one diagram and another is useful.

- What visual variables are being used to present data?
- What attributes of the networks are being shown and how?
- What benefit do these visualisations bring?

14. End of topic

Well done on reaching the end of **Topic: Visualising relational and network data**.



Reflect

Most of these relational and network data visualisations rely on using some visual or spatial metaphor for the abstract relationship. For instance, node-link diagrams use a line to show two entities are connected, while Euler diagrams use spatial containment to show set containment.

The use of such visual and spatial metaphors means that the visual representations are intuitive and easy to understand. However, these metaphors may not scale well to larger data sets. For larger data sets, matrix-based representations like the adjacency matrix for networks are often better. Their main disadvantage is that they are unfamiliar to most readers.

Next

You may now leave this topic book and return to this week's page. Talk to your peers and instructors if you have any thoughts or questions about the concepts, methods and examples mentioned related to this topic.